Michael Skinner

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Experience

Bell Canada

Cloud Automation Developer (Co-op) — *Toronto, ON* · **May 2024 - Aug 2024**

- Developed event-driven AWS Lambda pipelines using Python supporting large-scale device synchronization and backend workflows.
- Designed CI/CD validation gates preventing defective deployments and improving system reliability.
- Implemented structured logging and monitoring on Linux/Unix ensuring deterministic event processing.
- Investigated and resolved provisioning latency, configuration drift, and API inconsistencies across distributed systems.

Automation Intern (Co-op) — *Toronto, ON* · **May 2023 - Aug 2023**

- Developed systematic evaluation frameworks measuring extraction accuracy, consistency, and error rates for a Google Cloud–based OCR automation system extracting structured data from unstructured documents.
- Investigated failure modes in OCR outputs, diagnosed root causes, and designed regression test suites verifying extraction accuracy across diverse formats and edge cases.
- Implemented defensive validation checks preventing inconsistent downstream data ingestion.
- Built automated provisioning workflows using Python, SQL stored procedures, and REST APIs.

Superior Computer Solutions

Security Compliance Specialist (Co-op) — *Toronto, ON* · **May 2025 - Aug 2025**

- Conducted ISO/IEC 27001:2022 gap assessments and managed PCI-DSS SAQ-D audit evidence.
- Reviewed SOC 2 / ISO 27001 vendor reports and supported internal risk assessments.
- Strengthened expertise in secure system design, auditability, and integrity controls for sensitive data environments.

Projects

Autonomous Vehicle System (PX4 + Embedded + Distributed Control) — **Jan 2026 - Present**

- Developed an autonomous vehicle using PX4 on FMUK66 with NuttX RTOS for real-time embedded control and system coordination.
- Designed modular applications using uORB publish–subscribe messaging and MAVLink communication for distributed system integration.
- Integrated a Raspberry Pi for high-level sensor processing, enabling camera and ultrasonic-based obstacle avoidance.
- Implemented real-time motor control (ESC + servo) with feedback loops, focusing on reliability, timing constraints, and hardware–software integration.

Portable Ultrasound Device (Capstone) — **Sep 2025 - Present**

- Developed embedded firmware for high-speed ADC data acquisition on MCU platforms under real-time constraints.
- Designed and implemented real-time signal processing pipelines including filtering, envelope detection, and scan conversion.
- Integrated machine learning inference for automated detection of clinically relevant patterns from ultrasound data.
- Optimized end-to-end processing pipeline to maintain low-latency performance for real-time imaging and feedback.
- Engineered reliable data flow between hardware and software layers, ensuring signal integrity and robust system performance.

Optimization Research — **Sep 2025 - Dec 2025**

- Implemented AVX2/AVX512-vectorized sparse matrix kernels using cache-aware tiling and sparsity-driven optimizations.
- Developed OpenCL GPU kernels leveraging thread-level parallelism and memory coalescing for large sparse workloads.
- Tuned work-group sizing and kernel occupancy while using asynchronous command queues to maximize device throughput.
- Profiled CPU/GPU memory hierarchies to eliminate bottlenecks and proposed heterogeneous multi-core scaling strategies.

Software Defined Radio System — **Jan 2024 - May 2024**

- Implemented real-time IQ sampling pipelines for RF signal acquisition and processing using GNU Radio.
- Optimized FFT-based spectral analysis for efficient frequency-domain signal reconstruction.
- Designed and implemented custom protocol decoding logic with structured packet parsing and validation.
- Integrated C++/Python processing modules for low-latency streaming and real-time debugging.

Technical Skills

- **Programming**: Python, C, C++, Java, SQL, JavaScript
- **Systems & Embedded**: Distributed systems, real-time systems, Linux/Unix, CI/CD pipelines, Firmware, SPI/I2C/UART
- **Cloud & Automation**: AWS Lambda, Google Cloud, API development
- **Security & Compliance**: ISO 27001, PCI-DSS, SOC 2
- **Tools**: Git, Jira, FreeRTOS, GNURadio

Education

McMaster University — **Sep 2021 - Apr 2026**

B.Eng, Computer Engineering — *Hamilton, ON*